Solving the eigenvalue problem $\boldsymbol{H\psi} = E\psi$; $\quad \boldsymbol{H} = \begin{pmatrix} E_a & V_{ab} \\ V_{ba} & E_b \end{pmatrix}$; $\quad V_{ab} = V_{ba}^* = Ve^{-i\chi}$

Characteristic equation : $\begin{vmatrix} E_a - \lambda & V_{ab} \\ V_{ba} & E_b - \lambda \end{vmatrix} = 0$

eigenvalues : $\lambda \equiv E_\pm = \frac{1}{2}(E_a + E_b) \pm \frac{1}{2}\sqrt{(E_a - E_b)^2 + 4V^2} = E_m \pm \frac{\Delta}{2}$

Take E_m as a constant reference, then $E_\pm = \pm\frac{\Delta}{2}$

use $\tan 2\theta = \frac{2V}{E_a - E_b}; 0 < \theta < \frac{\pi}{2}$

then $\Delta^2 = \frac{4V^2}{\tan^2 2\theta} + 4V^2 = 4V^2\left(\frac{\cos^2 2\theta}{\sin^2 2\theta} + 1\right) = \frac{4V^2}{\sin^2 2\theta}$

$\therefore E_\pm = E_m \pm \frac{V}{\sin 2\theta}$

Consider E_+: $\begin{pmatrix} E_a - E_+ & V_{ab} \\ V_{ba} & E_b - E_+ \end{pmatrix}\begin{pmatrix} c_a \\ c_b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Let us simplify the matrix.

$$\begin{pmatrix} E_a - E_+ & V_{ab} \\ V_{ba} & E_b - E_+ \end{pmatrix} = \begin{pmatrix} E_a - E_m - \frac{\Delta}{2} & V_{ab} \\ V_{ba} & E_b - E_m - \frac{\Delta}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2}(E_a - E_b) - \frac{\Delta}{2} & V_{ab} \\ V_{ba} & -\frac{1}{2}(E_a - E_b) - \frac{\Delta}{2} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{V}{\tan 2\theta} - \frac{V}{\sin 2\theta} & Ve^{-i\chi} \\ Ve^{i\chi} & -\frac{V}{\tan 2\theta} - \frac{V}{\sin 2\theta} \end{pmatrix} = V\begin{pmatrix} -\frac{1-\cos 2\theta}{\sin 2\theta} & e^{-i\chi} \\ e^{i\chi} & -\frac{1+\cos 2\theta}{\sin 2\theta} \end{pmatrix}$$

now, $\frac{1-\cos 2\theta}{\sin 2\theta} = \frac{\sin^2\theta}{\sin\theta\cos\theta} = \frac{\sin\theta}{\cos\theta}$ and $\frac{1+\cos 2\theta}{\sin 2\theta} = \frac{\cos^2\theta}{\sin\theta\cos\theta} = \frac{\cos\theta}{\sin\theta}$

then the eigenvalue problem becomes :

$$\begin{pmatrix} -\frac{\sin\theta}{\cos\theta} & e^{-i\chi} \\ e^{i\chi} & -\frac{\cos\theta}{\sin\theta} \end{pmatrix}\begin{pmatrix} c_a \\ c_b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \text{ or, } \begin{matrix} c_a = \cos\theta e^{-\frac{i\chi}{2}} \\ c_b = \sin\theta e^{\frac{i\chi}{2}} \end{matrix}$$

We can similarly find out the coefficients for the other eigenvalue, E_-: $\begin{matrix} c_a = -\sin\theta e^{-\frac{i\chi}{2}} \\ c_b = \cos\theta e^{\frac{i\chi}{2}} \end{matrix}$

$\therefore$ in the uncoupled basis $\{|\phi_a\rangle, |\phi_b\rangle\}$

$$\begin{matrix} |\psi_+\rangle = \cos\theta e^{-\frac{i\chi}{2}}|\phi_a\rangle + \sin\theta e^{\frac{i\chi}{2}}|\phi_b\rangle \\ |\psi_-\rangle = -\sin\theta e^{-\frac{i\chi}{2}}|\phi_a\rangle + \cos\theta e^{\frac{i\chi}{2}}|\phi_b\rangle \end{matrix}$$